

Damian Kim

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EDUCATION

Duke University

Expected Graduation: May 2027

Bachelor of Science | Double Major: Computer Science and Mathematics

GPA: 3.8

Relevant Coursework: Theory & Algorithms of Machine Learning, Design & Analysis of Algorithms, Applied Stochastic Processes, Bayesian & Modern Statistics, Statistical Computing, Mathematical Finance, Algorithmic Trading, Computer Vision, Probability, Regression Analysis, Data Structures & Algorithms, Database Systems, Computer Systems, Discrete Mathematics

WORK EXPERIENCE

Golfie (Golf Simulation Software)

Durham, NC

Founder and Software Engineer

May 2026 - Present

- Architected the full pipeline for simulating and reconstructing golf ball trajectories from dual-camera iPhone video
- Built a **custom physics engine** in Python/NumPy using RK4 numerical integration with gravity, drag, and Magnus lift
- Created a computer vision stereo pipeline (**OpenCV**) for calibration, triangulation, and launch parameter estimation
- Developed an audio-based video sync system using **FFmpeg** and cross-correlation to align independent video streams.
- Engineered a 2D ball-detection algorithm using median background modeling, contour convex hulls, circularity metrics
- Formulated golf ball launch speed, angle, and spin estimation algo using SciPy non-linear least-squares fitting
- Generated an interactive 3D driving range (w **React**, **TypeScript**, **Three.js**) to render simulated ball trajectory curves

Duke University

Durham, NC

Teaching Assistant (Discrete Math)

January - April 2026

- Taught weekly recitations, ran office hours, graded and edited exams for a Discrete Mathematics course at Duke

Mayo Clinic

Rochester, MN

SWE Intern (AI FAST team)

June 2023 - Aug 2024

- Engineered scalable backend integrations of proprietary **ML APIs** with data processing pipelines using **Python** and **REST** frameworks, enabling efficient deployment and versioning of AI models in production environments
- Automated EDA by refactoring **Jupyter notebooks** into **Python scripts** for an automated integrated research pipeline
- Assisted with the team's migration from **Git** to **Azure DevOps**, redesigning source control and **CI/CD** workflows
- Created cloud-based data query pipelines on **GCP** using **BigQuery** and **Vertex AI**, replacing legacy on-prem **SQL**
- Implemented **frequency-based caching** and data clustering strategies for a 2+ TB radiology imaging dataset, significantly optimizing repeated query execution to reduce average scan volume by 20x and latency by 15-70%

NASA Langley

Hampton, VA

Summer Research Mentee

June - August 2021

- Increased void analysis accuracy by creating and testing a **Keras Tensorflow**-based convolutional neural network (**CNN**) trained on cross-sectional images of polymer composites as compared to manual ImageJ analysis
- Identified failure modes and proposed AI architecture improvements for broader analysis applications in materials

PROJECTS

Juno (VoIP app)

June 2026

- Built a low-latency voice calling app using **Agora RTC** to deliver **sub-200ms** (US to China) latency audio and video
- Engineered a screen-sharing system with **1080p 30fps** video quality and a texting chat with client-side persistence
- Developed an AI-assisted live subtitling system via **HTML5 MediaRecorder** over **WebSockets**, uses Deepgram to render SST input, translate with Google Cloud Translation, and TTS output through the same WebSocket
- Added Wordle-type co-op game inside calls utilizing shared **Zustand** store states to coordinate player guesses

Sate App

June - July 2025

- Built a full-stack **Flask** app with **MongoDB** database backend enabling restaurant recs from Tinder-esque swiping
- Tested a weighted group recommendation model incorporating negative user feedback for preference optimization, achieving ~65% higher consensus accuracy in survey testing compared to baseline random recommendation

SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, R, HTML/CSS

Frameworks and Tools: NumPy, SciPy, Pandas, Scikit-Learn, React, Next.js, Flask, FastAPI, OpenCV, TensorFlow/Keras